

Endothermic and exothermic reactions

Learning outcomes

By the end of this section you should be able to:

- (R1.1.2) Identify the direction of energy transfer between system and surroundings for endothermic and exothermic reactions.
- (R1.1.2) Explain the temperature change (decrease or increase) that accompanies endothermic and exothermic reactions, respectively.
- (R1.1.3) Sketch and interpret energy profiles for endothermic and exothermic reactions.

It is common for people to associate chemical reactions with the release of energy. Usually this is because an increase in temperature of the surroundings is detected during many chemical reactions. Many reactions do, however, absorb energy. Arguably one of the most important chemical reactions on the planet requires the absorption of energy to convert carbon dioxide and water into carbohydrates and oxygen. What do you think this reaction might be?

Photosynthesis!



Photosynthesis: one of the most important reactions on the planet.

Credit: Sarayut Thaneerat, Getty Images

How might we describe reactions in terms of how energy is transferred? How might we expect the temperature to change as a result of energy transfer? How might we illustrate the energy changes of a chemical reaction graphically?

Direction of energy transfer

We can categorise chemical reactions in terms of their direction of energy transfer. Reactions that release energy to the surroundings are referred to as exothermic whereas reactions that absorb energy from the surroundings are referred to as endothermic. Energy transferred is typically in the form of heat but may also be electricity or light. **Figure 1** below illustrates the transfer of energy between system and surroundings for endothermic and exothermic processes.

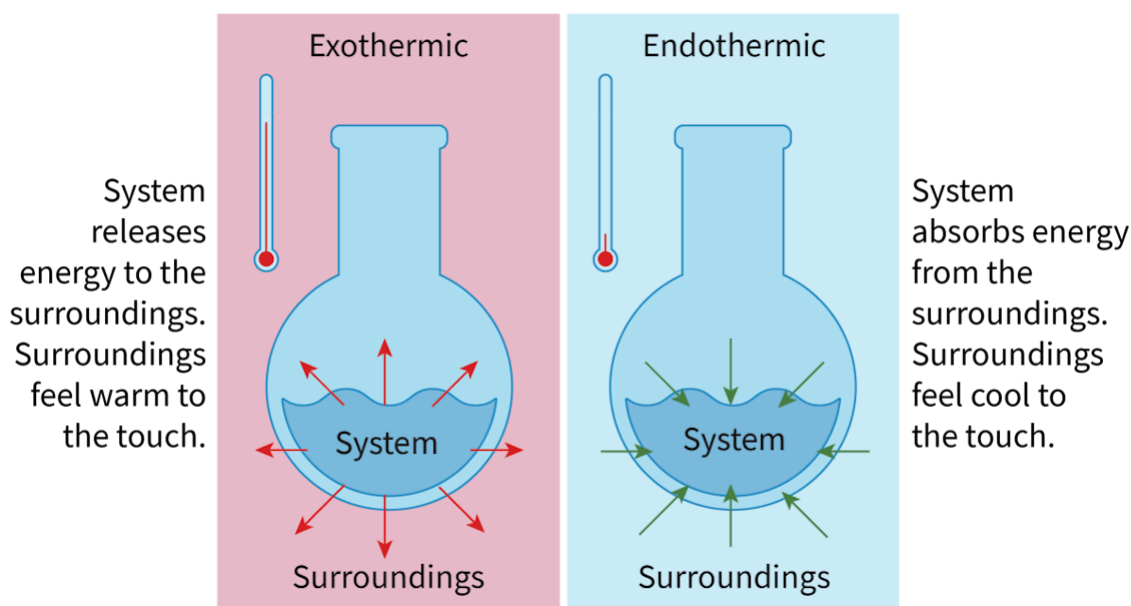


Figure 1. Transfer of energy during exothermic and endothermic processes.

 More information for figure 1

The diagram is divided into two sections, with each section depicting a flask and a thermometer. The left section is labeled 'Exothermic' and shows a flask with arrows pointing outward from the system, indicating that energy is being released to the surroundings. The text beside it states, 'System releases energy to the surroundings. Surroundings feel warm to the touch.' The thermometer beside the flask shows a high temperature.

The right section is labeled 'Endothermic' and shows a flask with arrows pointing inward to the system, indicating that energy is being absorbed from the surroundings. The text beside it reads, 'System absorbs energy from the surroundings. Surroundings feel cool to the touch.' The thermometer beside this flask shows a low temperature. Each section illustrates the direction of thermal energy transfer, affecting the temperature of the surroundings.

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Processes that are endothermic and exothermic

When you think of the physical changes of states covered in [section S1.1.2a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/>) can you identify which ones would be endothermic and which would be exothermic? Think about which processes involve an absorption of energy versus a release of energy. Try the drag and drop **Interactive 1** to classify the changes of state as endothermic or exothermic.

Interactive 1. Endothermic or Exothermic Processes

Can you think of any examples of chemical reactions that are always endothermic or exothermic? Think about the combustion engines in our cars or the process of plants making glucose from carbon dioxide and water. While many reactions will be exothermic or endothermic depending on the identity of the reactants, many types of reaction will always release or absorb energy. **Table 1** summarises some of these processes.

Table 1. Chemical and physical changes that are **always** endothermic

	Exothermic		Er
	Type of reaction or process	Example	Type of reaction or process
Chemical change	Combustion	$\text{CH}_4(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$	Photosynthesis
Chemical change	Neutralisation	$\text{NaOH}(\text{aq}) + \text{HCl}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$	<u>Thermal decomposition</u>
Physical change	Freezing	$\text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{O}(\text{s})$	Melting
Physical change	Condensation	$\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$	Evaporation/boiling
Physical change	Deposition	$\text{H}_2\text{O}(\text{g}) \rightarrow \text{H}_2\text{O}(\text{s})$	Sublimation

Effects on temperature

How do you expect the temperature to change during exothermic and endothermic reactions? When we measure the temperature of most reactions, are we measuring the temperature of the system or the surroundings? Let's consider an exothermic acid–base reaction in **Figure 2**.

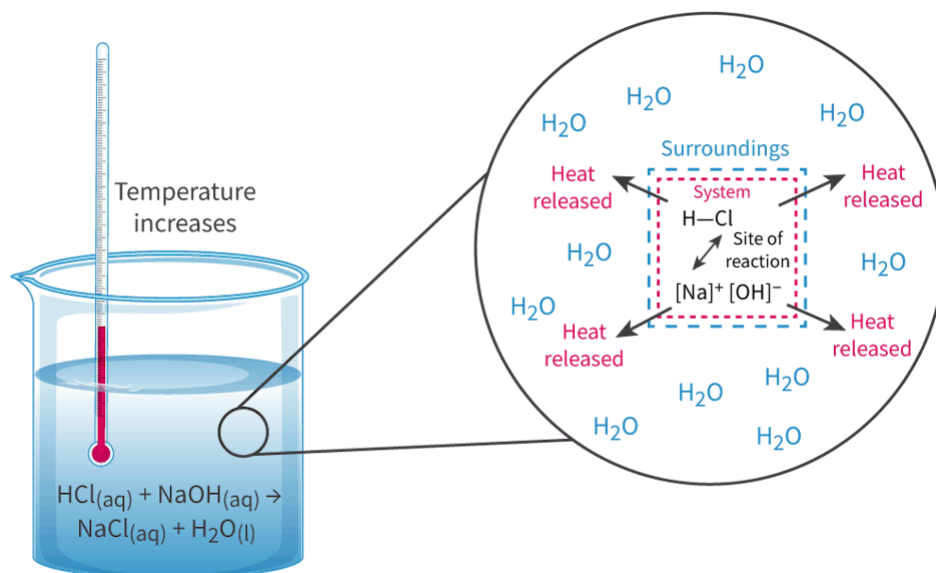


Figure 2. A close look at the system and surroundings during an exothermic acid—base reaction.

 More information for figure 2

The image is a diagram illustrating an exothermic acid-base reaction. On the left side is a beaker filled with a liquid, depicted to represent the aqueous solution involving the reactants HCl(aq) and NaOH(aq) . Above the liquid in the beaker, a thermometer shows the temperature increasing, indicative of the exothermic reaction.

To the right, a magnified view highlights the system and surroundings concept. Inside a box labeled "System," indicated by a dashed line, is where the reactants HCl and NaOH interact. Arrows point outward from the "System" to an area labeled "Surroundings," depicting the release of heat energy.

The surroundings are shown with multiple " H_2O " labels, representing the solvent, water. This visualization emphasizes how heat from the reaction in the system is transferred to the water, increasing the temperature of the surroundings. The phrase "Heat released" accompanies each arrow pointing away from the "System." This setup serves to establish a clear distinction between the system and the surroundings in the context of chemical reactions.

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During most chemical reactions, when we measure the temperature we are often measuring the temperature of the surroundings. In **Figure 2**, we can see that an acid–base reaction involves an aqueous acid reacting with an aqueous base. Section S1.1.1a ([https://app.kognity.com/study/app/preview-p/sid-426-cid-](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/endothermic-and-exothermic-reactions-id-45282/print/)

[753301/book/elements-compounds-and-mixtures-id-43065/](#)) shows that aqueous solutions involve a solute (HCl or NaOH in this example) dissolved in water as the solvent. When the reactants, HCl and NaOH, react, the site where these reactants react is the system. Everything around this site is a part of the surroundings. This means that the solvent, water, would act as the surroundings.

When the **exothermic** reaction between HCl and NaOH **releases energy** in the form of heat, the water molecules surrounding the site of reaction increase in kinetic energy. This results in an **increase in temperature** of the aqueous solution that we can measure using a thermometer.

What about endothermic processes? Have you ever felt an alcohol, such as that in hand sanitiser, on the surface of your skin? What does your skin feel like afterwards? The alcohol used in sanitiser alcohol is often used as an antiseptic to clean the skin. After use, the skin usually feels cool. This is due to the evaporation of the alcohol in the hand sanitiser!

As the alcohols used in hand sanitiser have a relatively low boiling point, the heat energy from your hands (surroundings) can be absorbed by the alcohol in the sanitiser (the system) leading to evaporation. The kinetic energy of molecules in your hands is slightly lowered due to this loss of heat energy resulting in a slightly lower temperature. This process is illustrated in **Figure 3**. The **endothermic** process of the evaporation of alcohol **absorbs energy** from the surroundings resulting in a **decrease in temperature** of the surroundings.

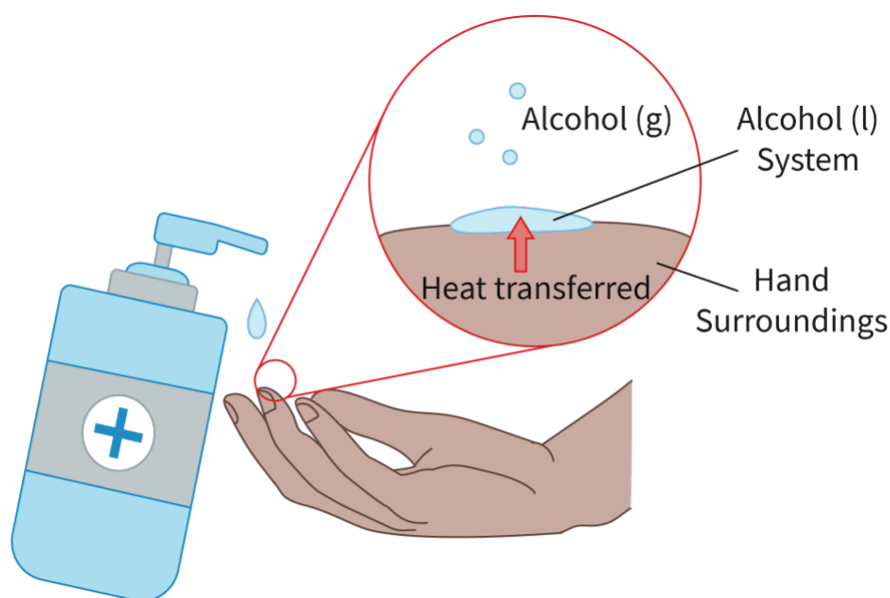


Figure 3. Hand sanitiser containing alcohol that easily evaporates in our hands.

 More information for figure 3

The image is a diagram illustrating the evaporation process of hand sanitiser. It shows a hand with sanitiser being applied from a bottle. There is an enlarged view of the hand where a liquid droplet, labeled "Alcohol (l)", is shown on the palm, indicating the 'system'. Above it, there are smaller circles representing gaseous alcohol, labeled "Alcohol (g)", showing the evaporation process. An arrow labeled "Heat transferred" points from the hand to the droplet, representing the heat energy transfer from the hand to the alcohol, resulting in the evaporation. The hand is labeled as "Surroundings", and the concept of alcohol evaporating from liquid to gas is visually depicted with annotations.

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Energy profiles

Section R1.1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/energy-transfer-in-chemical-reactions-id-43469/>) considers different forms of energy that exist. Recall that one type of potential energy is chemical potential energy. Chemical potential energy is the energy stored in the chemical bonds between particles. The quantity of stored energy depends on the arrangement and type of bonding that exists between particles at the atomic level.

We can represent the chemical potential energy of products and reactants from endothermic and exothermic reactions graphically using an energy profile diagram. These energy profiles show the relative stability of reactants and products by considering the relative chemical potential energy each has. Look at the energy profile diagrams in **Figure 4**.

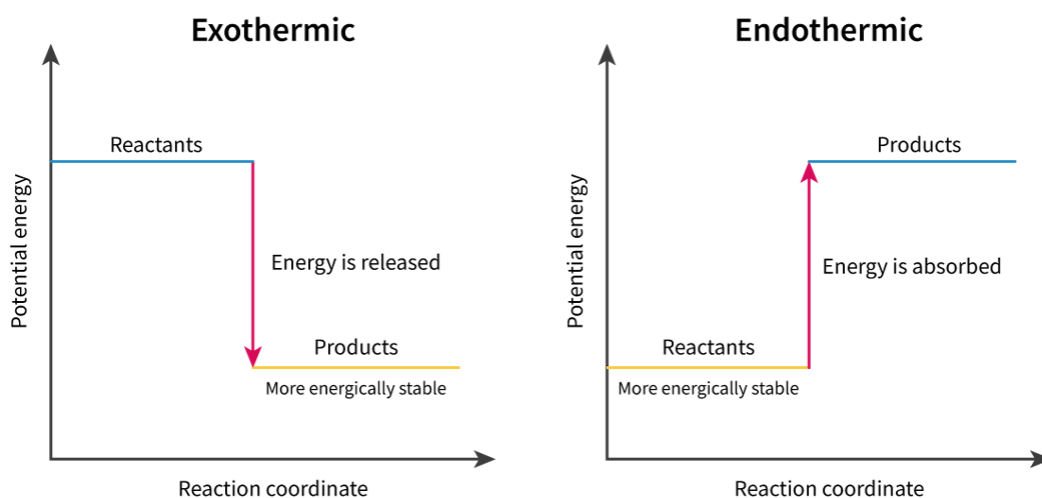


Figure 4. Energy profile diagrams for endothermic and exothermic reactions.

 More information for figure 4

The image consists of two separate diagrams: an exothermic reaction on the left and an endothermic reaction on the right.

In the exothermic diagram: - The X-axis represents the progression of the reaction, while the Y-axis shows energy level. - The curve starts high, representing reactants with high potential energy, and drops sharply downwards, ending with products at a lower energy level. This indicates energy release.

In the endothermic diagram: - The X-axis also represents the reaction progress and the Y-axis shows energy. - This curve starts low, depicting reactants with lower potential energy, and rises to show that products have higher energy, representing absorbed energy.

Both diagrams illustrate the shifts in energy between reactants and products, showing how potential energy aligns with stability.

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You can see in **Figure 4** how exothermic processes show a release of energy with the products ending with a lower potential energy than the reactants. You can see here that stability and potential energy are interrelated. Similarly, in **Figure 4**, you can see how endothermic processes show an absorption of energy with the reactants having greater stability due to the lower potential energy. A lower potential energy leads to greater stability. You can think of it as being similar to a ladder. The lower on the ladder you stand the greater stability you have. The higher a ladder you stand the less stable you are.

Try the drag and drop in **Interactive 2** to match all missing terms to the correct places in the two energy profile diagrams.

Interactive 2. Energy profile diagrams.

 More information for interactive 2

A drag-and-drop interactivity features two energy profile diagrams with missing terms. Users need to drag and drop the correct terms in the corresponding drop zones. Both diagrams plot potential energy on the y-axis versus reaction coordinate on the x-axis.

In the first diagram, labeled drop zone, the reactants have a high potential energy and are converted into products with lower potential energy. Here, the energy is drop zone. Drop zones are more stable.

In the first diagram, labeled drop zone, the reactants have low potential energy and are converted into products with high potential energy. Here, the energy is drop zone. Drop zones are more stable.

The drag-and-drop options are as follows:

Released

Reactants

Exothermic

Products

Absorbed

Endothermic

This interactivity helps users differentiate between exothermic and endothermic reactions based on energy profile diagrams.

Read below for the solution:

In the first diagram, labeled exothermic, the reactants have a high potential energy and are converted into products with lower potential energy. Here, the energy is released. Products are more stable.

In the second diagram, labeled endothermic, the reactants have a low potential energy and are converted into products with high potential energy. Here, the energy is absorbed. Reactants are more stable.

Activity

- **IB learner profile attribute:** Principled
- **Approaches to learning:**
 - Research skills — Using a single, standard method of referencing and citation
 - Thinking skills — Being curious about the natural world
- **Time required to complete activity:** 30—45 minutes
- **Activity type:** Individual activity

Research a physical process or chemical reaction with a notable energy transfer that is applicable to everyday life. Some examples you could choose include:

- sodium ethanoate (sodium acetate) heat packs
- hand warmers
- instant cold packs
- MREs (Meals, Ready to Eat or ration packs)
- cellular respiration
- formation of snow
- making ice cream.

Create an infographic (using a digital tool such as [Canva](https://www.canva.com/)  (<https://www.canva.com/>)) showing what you have learned. Your infographic should include the following:

- Identification of the chemical or physical process that is occurring.
- Classification of the process as either endothermic or exothermic.
- Diagram labelling system, surroundings and direction of energy transfer.
- Description of expected temperature changes.
- Energy profile diagram.

- Why is energy transfer important in this real-world application.
- 3—5 interesting facts.
- Proper citations for any references used.